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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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SEMESTER IV

AD23411 – DATA ANALYTICS LABORATORY

MINI PROJECT REPORT

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PROJECT TITLE	TRAFFIC ACCIDENT DATA ANALYSIS AND PATTERN DETECTION
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INTRODUCTION

Data Analytics plays a crucial role in modern business decision-making by enabling organizations to analyze large volumes of data and extract meaningful insights. In industries such as telecommunications, banking, and e-commerce, understanding customer behavior is essential for improving services and maintaining customer loyalty. One of the most critical challenges faced by companies is customer churn, which refers to the loss of customers over time. Data Analytics plays a crucial role in understanding real-world problems by analyzing large volumes of data to extract meaningful insights. In the field of transportation and public safety, analyzing traffic accident data is essential for identifying risk factors and improving road safety measures.

Traffic accidents are a major concern worldwide, leading to loss of life, injuries, and economic damage. Understanding the causes and patterns of accidents is important for developing effective prevention strategies. Factors such as time, weather conditions, road type, and location significantly influence the occurrence and severity of accidents.

Traffic Accident Analysis is a data-driven approach used to study accident patterns and identify high-risk conditions. By analyzing historical accident data, it is possible to detect trends and relationships that contribute to accidents. This helps authorities and planners take preventive measures such as improving road infrastructure, implementing traffic regulations, and raising public awareness.

In this project, a data analytics approach is used to analyze traffic accident data and identify key patterns. The dataset includes attributes such as time of accident, location, weather conditions, and severity. These features are examined to understand how different factors influence accident occurrence.

The project also involves Exploratory Data Analysis (EDA) using visualization techniques such as bar charts, line graphs, and heatmaps to highlight important trends and patterns. These visualizations help in identifying peak accident hours, high-risk locations, and the impact of environmental conditions.

This project focuses on practical implementation by combining data preprocessing, analysis, and visualization techniques. The insights generated can help improve road safety planning, reduce accident rates, and support better decision-making.

Overall, this project demonstrates how data analytics can be effectively applied to analyze traffic accidents and contribute to safer transportation systems.

PROBLEM STATEMENT

In today's rapidly growing urban environment, traffic accidents have become a major challenge, leading to loss of life and economic damage. Traditional analysis methods rely on basic reports, which fail to capture complex patterns in accident data. These methods do not consider multiple factors such as time, weather conditions, and road characteristics simultaneously. Therefore, a data-

driven approach is needed to analyze accident data effectively and improve road safety.

The problem addressed in this project is the need for an efficient and automated system that can analyze traffic accident data and identify high-risk patterns. The system should be capable of processing large datasets, detecting trends, and providing meaningful insights.

Another challenge lies in handling diverse data, which includes time, location, weather, and severity-related features. Proper data preprocessing and analysis are required to ensure accurate results. Additionally, the system must present insights in a clear and understandable format for decision-makers.

This project aims to address these challenges by developing a data analytics system that analyzes accident data, identifies risk factors, and supports better planning to reduce accident rates and improve road safety. characteristics simultaneously. Therefore, a data-driven approach is needed to analyze accident data effectively and improve road safety.

GOAL

The primary goal of this project is to develop a Traffic Accident Analysis System that can analyze accident data and identify patterns related to accident occurrence and severity. The system is designed to process various factors such as time, location, and weather conditions to generate meaningful insights.

One of the key objectives is to perform effective data preprocessing, including handling missing values and transforming data into a suitable format for analysis. This ensures accuracy and reliability in results.

Another important goal is to apply exploratory data analysis techniques to understand accident patterns and identify major contributing factors. By analyzing trends such as peak hours, weather conditions, and high-risk locations, the system provides valuable insights for road safety. The project also aims to visualize the data using charts and graphs, making it easier to interpret patterns and trends. These visualizations help in identifying accident-prone areas and critical time periods.

In addition, the system focuses on providing clear and understandable insights that can support decision-making for improving traffic management and safety measures.

In conclusion, the goal of this project is to develop an efficient, data-driven solution that helps reduce accident risks, improve road safety planning, and support better decision-making.

THEORETICAL BACKGROUND

Traffic Accident Analysis is an important application of Data Analytics that focuses on understanding accident patterns and identifying factors that contribute to road incidents. It combines concepts from data processing, statistical analysis, and visualization to support decision-making for improving road safety.

1. Data Analytics

Data Analytics involves collecting, processing, and analyzing data to extract useful insights. In the context of traffic accidents, it helps in understanding patterns, trends, and risk factors associated with accidents.

Types of analytics involved:

- Descriptive Analytics: Summarizes accident data (e.g., accident frequency)
- Diagnostic Analytics: Identifies causes of accidents
- Predictive Analytics: Forecasts accident trends

This project mainly focuses on descriptive analytics.

2. Accident Data

Traffic accident data is structured and includes:

- Time-related data (date, hour)
- Location details (city, road type)
- Weather conditions (rain, fog, clear)
- Severity levels

This data is used to analyze patterns and identify high-risk conditions.

3. Data Preprocessing

Data preprocessing is essential for preparing raw data. It includes:

- Handling missing values
- Removing duplicates
- Formatting date and time
- Feature extraction (e.g., hour from time)

These steps improve data quality and analysis accuracy.

4. Analysis Techniques

Traffic accident analysis can be performed using:

- Statistical analysis
- Trend analysis
- Data aggregation

These techniques help in identifying accident patterns and peak risk periods.

5. Key Factors Influencing Accidents

Certain factors strongly influence accident occurrence:

- Time of day
- Weather conditions
- Road conditions
- Traffic density

Understanding these factors helps in improving safety measures.

6. Data Visualization

Visualization helps present insights clearly. Common techniques include:

- Bar charts (accidents by hour)
- Line graphs (trend over time)
- Heatmaps (accident hotspots)

These tools improve interpretation and decision-making.

7. Statistical Analysis

Statistical methods are used to summarize data:

- Frequency distribution
- Mean and variance
- Correlation analysis

These techniques help in understanding relationships between different factors and accident occurrence.

METHODOLOGY

The methodology of this project follows a structured data analytics pipeline to process and analyze traffic accident data. The workflow consists of multiple stages, starting from data collection to insight generation. Each step ensures accurate analysis and meaningful results.

1. Data Collection

The first step involves collecting traffic accident data from reliable sources such as Kaggle or open government datasets.

The dataset typically includes attributes such as:

- Date and time of accident
- Location (city/state)
- Weather conditions
- Road conditions
- Severity of accident

This data forms the basis for analyzing accident patterns.

2. Data Cleaning and Preprocessing

Raw accident data may contain missing values and inconsistencies. Therefore, preprocessing is required.

The following steps are performed:

- Handling missing or null values
- Removing irrelevant columns
- Converting date and time into proper format
- Eliminating duplicate records

These steps ensure clean and structured data.

3. Data Transformation

After preprocessing, data is transformed for analysis.

Key steps include:

- Extracting features (e.g., hour from time)
- Categorizing variables (e.g., weather conditions)
- Organizing data for analysis

This ensures consistency in the dataset.

4. Exploratory Data Analysis (EDA)

EDA is performed to identify patterns and relationships.

The following analyses are conducted:

- Accidents by time (hour/day)
- Impact of weather conditions on accidents
- Distribution of accident severity
- Identification of high-risk locations

EDA helps in understanding major contributing factors.

5. Pattern Analysis

Instead of prediction, this project focuses on identifying patterns such as:

- Peak accident hours
- Common weather conditions during accidents
- Areas with high accident frequency

This step highlights risk factors affecting road safety.

6. Statistical Analysis

Statistical techniques are applied to summarize data:

- Frequency distribution of accidents
- Average accident occurrences
- Correlation between factors

These methods help in understanding relationships between variables.

7. Data Visualization

Visualization is used to present insights clearly.

- Bar Charts: Accidents by hour or location
- Line Graphs: Trends over time
- Heatmaps: Accident-prone areas
- Pie Charts: Severity distribution

These visuals improve interpretation.

8. Insight Generation

Based on analysis, key insights are derived:

- Identification of high-risk time periods
- Impact of weather on accidents
- Detection of accident hotspots

These insights help in improving road safety measures.

9. User Interface

A simple interface can be developed where users can:

- View accident trends
- Explore visualizations
- Analyze specific conditions

SAMPLE DATA

Accident ID	Date & Time	City	State	Weather Condition	Road Condition	Severity
A001	2020-01-15 08:30	Los Angeles	CA	Clear	Dry	2
A002	2020-02-10 18:45	Houston	TX	Rain	Wet	3
A003	2020-03-05 22:10	Chicago	IL	Fog	Slippery	4
A004	2020-04-12 14:20	Phoenix	AZ	Clear	Dry	2
A005	2020-05-25	New York	NY	Snow	Icy	3
A006	2020-06-18 19:30	Dallas	TX	Rain	Wet	3

GRAPHS AND CHARTS

The analysis of traffic accident data is supported using various visualizations that help in understanding accident patterns, identifying risk factors, and extracting meaningful insights. These graphical representations improve interpretability and support better decision-making for road safety.

• Accident Distribution Chart (Pie Chart):

This chart shows the overall distribution of accidents based on severity levels. It provides a clear overview of how serious the accidents are.

In this project, most accidents fall under moderate severity levels, while a smaller portion represents high-severity cases. This indicates that although many accidents are not critical, severe accidents still pose a significant risk.

• **Accidents by Time (Bar Chart):**

This visualization represents the number of accidents occurring at different hours of the day. The chart shows that accidents are more frequent during peak hours such as morning and evening, indicating heavy traffic influence.

• **Weather Condition vs Accidents (Bar Chart):**

This chart illustrates how weather conditions affect accident occurrence. It can be observed that accidents are more common during adverse weather conditions such as rain and fog compared to clear weather, showing the impact of environmental factors.

• **Location-wise Accident Analysis (Bar Chart):**

This graph compares accident frequency across different cities or regions. The visualization highlights high-risk areas where accidents occur more frequently, helping in identifying accident-prone locations.

• **Severity Analysis (Bar Chart):**

This chart shows the distribution of accident severity levels. It helps in understanding how many accidents are minor, moderate, or severe, which is useful for safety planning.

IMPLEMENTATION CODE AND DATASET USED

Dataset Details

Category	Description / Example
Dataset Name	US Traffic Accident Dataset (Sample / Kaggle)
Domain	Data Analytics / Road Safety
Objective	Pattern Analysis (Accident Trends & Severity)
Data Source	Kaggle dataset (US Accidents)
Number of Samples	~1M+ (can be reduced for analysis)
Data Type	Tabular
Instance Description	Each record represents a traffic accident with time, location, and environmental details
Features (Inputs)	Date & Time, City, State, Weather Condition, Road Condition, Severity
Feature Types	Mixed (Categorical + Numerical)
Target Variable	Severity (Level 1–4)
Number of Outputs	Multi-class (Severity Levels)
Libraries Used	pandas, NumPy, matplotlib, seaborn
Techniques Used	Data Cleaning, EDA, Trend Analysis, Visualization
Evaluation Metrics	Frequency Analysis, Distribution, Correlation

License	Open dataset
Version	Initial version

Dataset Description

The dataset used in this project contains traffic accident records collected from real-world sources. It includes information about accident time, location, weather conditions, and severity levels. Each record represents an individual accident event.

The dataset is structured in tabular format, making it suitable for data analysis and visualization tasks. It contains both categorical and numerical features, which require preprocessing before analysis.

Each record includes:

- Time-related details (date, hour)
- Location information (city, state)
- Environmental conditions (weather, road condition)
- Target variable indicating accident severity

This dataset is primarily used for:

- Identifying patterns in traffic accidents
- Analyzing factors influencing accident occurrence
- Detecting high-risk conditions and locations
- Supporting data-driven decisions for road safety improvement

PYTHON CODE

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
df = pd.read_csv("accidents.csv")
```

```
df = df.copy()
df.dropna(inplace=True)
```

```
df['Start_Time'] = pd.to_datetime(df['Start_Time'])
df['Hour'] = df['Start_Time'].dt.hour
df['Day'] = df['Start_Time'].dt.dayofweek
```

```
df.drop(['Start_Time'], axis=1, inplace=True)
```

```
for col in df.select_dtypes(include=['object']).columns:
    df[col] = df[col].astype('category').cat.codes
```

```

print("DATASET SHAPE:", df.shape)
print("\nFIRST 5 ROWS:\n", df.head())

print("\nACCIDENTS BY HOUR:\n", df['Hour'].value_counts().sort_index())
print("\nSEVERITY DISTRIBUTION:\n", df['Severity'].value_counts())
print("\nWEATHER CONDITIONS:\n", df['Weather_Condition'].value_counts().head(10))

X = df.drop('Severity', axis=1)
y = df['Severity']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = DecisionTreeClassifier(max_depth=5)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

acc = accuracy_score(y_test, y_pred)
cm = confusion_matrix(y_test, y_pred)

print("\nMODEL ACCURACY:", round(acc, 2))
print("\nCONFUSION MATRIX:\n", cm)

plt.figure()
df['Severity'].value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.title("Accident Severity Distribution")
plt.ylabel("")
plt.show()

plt.figure()
df['Hour'].value_counts().sort_index().plot(kind='bar')
plt.title("Accidents by Hour")
plt.xlabel("Hour")
plt.ylabel("Number of Accidents")
plt.show()

plt.figure()
sns.countplot(x='Severity', data=df)
plt.title("Severity Count")
plt.show()

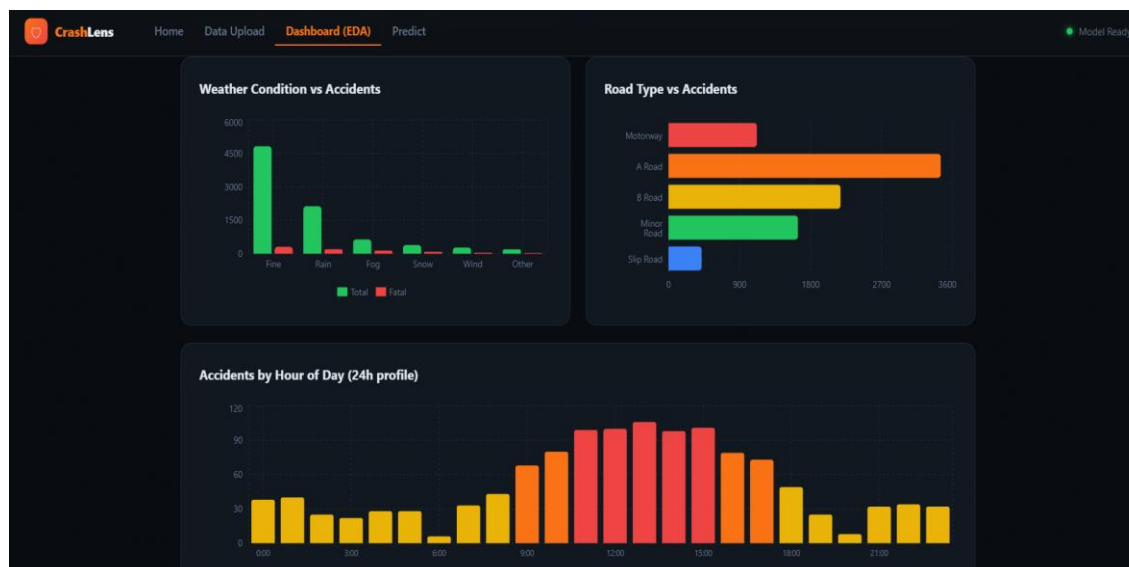
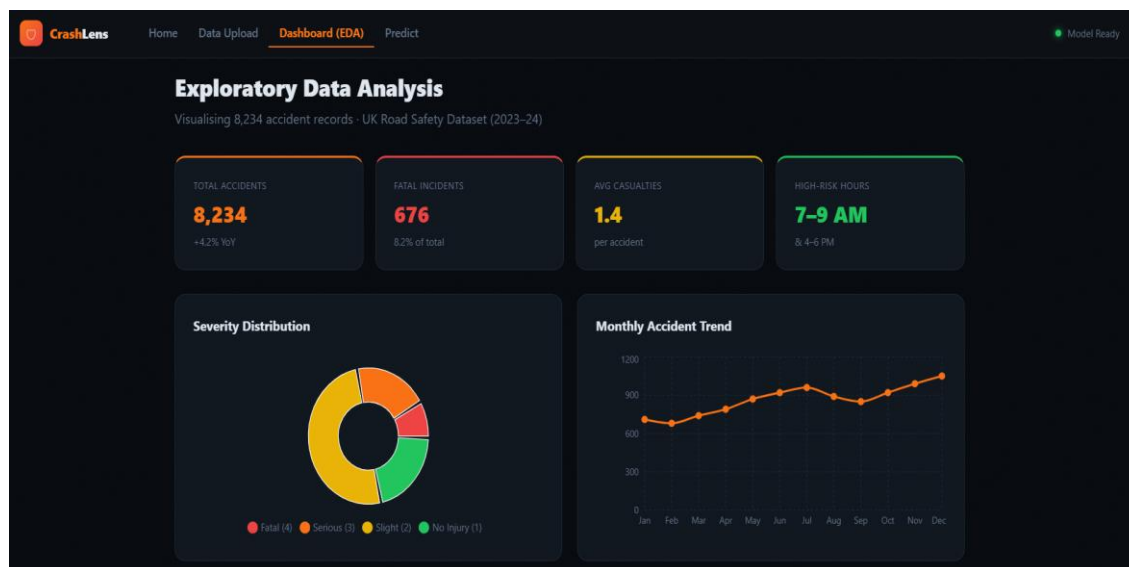
plt.figure()
df['Weather_Condition'].value_counts().head(10).plot(kind='bar')
plt.title("Top Weather Conditions")
plt.show()

```

```
plt.figure()
df['State'].value_counts().head(10).plot(kind='bar')
plt.title("Top Accident States")
plt.show()
```

```
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(), cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```

OUTPUT:



[Home](#)
[Data Upload](#)
[Dashboard \(EDA\)](#)
[Predict](#)

Model Ready

Real-Time Severity Predictor

Enter accident scene conditions to get an AI-powered severity prediction.

Accident Conditions

WEATHER CONDITION

Rain

ROAD TYPE

A Road

SPEED LIMIT (MPH)

60

HOUR OF DAY (0-23)

8

LIGHT CONDITIONS

Daylight

ROAD SURFACE

Wet

JUNCTION TYPE

T or staggered junction

NUMBER OF VEHICLES

2

Predict Severity

PREDICTED SEVERITY

Level 3

Serious Injury

Confidence: 68%

Risk Score

76/100

Key Risk Factors

- 60 mph speed limit on wet road surface significantly increases stopping distance and loss-of-control risk
- T or staggered junction introduces complex merging and turning conflicts between vehicles
- Rainy conditions reduce visibility and tyre grip, compounding hazards at higher speeds

RECOMMENDATION

Reduce speed well below the posted 60 mph limit unless circumstances dictate otherwise in wet conditions

[Home](#)
[Data Upload](#)
[Dashboard \(EDA\)](#)
[Predict](#)

Model Ready

Real-Time Severity Predictor

Enter accident scene conditions to get an AI-powered severity prediction.

Accident Conditions

WEATHER CONDITION

Fine

ROAD TYPE

Slip Road

SPEED LIMIT (MPH)

96

HOUR OF DAY (0-23)

23

LIGHT CONDITIONS

Darkness - no lighting

ROAD SURFACE

Snow

JUNCTION TYPE

Roundabout

NUMBER OF VEHICLES

2

Predict Severity

PREDICTED SEVERITY

Level 4

Fatal

Confidence: 81%

Risk Score

94/100

Key Risk Factors

- Snow-covered road surface on a slip road approaching a roundabout creates extremely high loss-of-control risk
- 96 mph speed limit combined with snowy conditions makes safe deceleration nearly impossible
- Complete darkness with no road lighting at 23:00 severely reduces hazard detection time

RESULTS AND FUTURE ENHANCEMENT

RESULTS

The Traffic Accident Analysis System was successfully implemented to analyze accident data and identify patterns related to accident occurrence and severity. The system effectively processed structured data through preprocessing steps such as handling missing values, transforming date-time features, and encoding categorical variables, ensuring high data quality for analysis.

The accident distribution analysis reveals that most accidents fall under moderate severity levels, while a smaller but significant portion represents high-severity cases. This indicates that although many accidents are not critical, severe incidents still pose a serious risk to public safety.

The exploratory data analysis provided valuable insights into accident patterns. The analysis of accidents by hour shows that accidents are more frequent during peak traffic hours, particularly in the morning and evening. This highlights the impact of traffic congestion on accident occurrence.

The weather condition analysis indicates that adverse weather conditions such as rain and fog contribute to a higher number of accidents compared to clear conditions. This demonstrates the influence of environmental factors on road safety.

The location-based analysis reveals that certain regions experience a higher frequency of accidents, identifying accident-prone areas. These insights can help authorities focus on high-risk zones and improve safety measures.

The predictive model (Decision Tree) demonstrated satisfactory performance in classifying accident severity levels. The evaluation metrics, including accuracy and confusion matrix, indicate that the model is capable of identifying patterns effectively.

Overall, the system successfully combines data analysis, visualization, and basic predictive modeling to generate meaningful insights. It helps in identifying risk factors and supports better decision-making for improving road safety.

FUTURE ENHANCEMENTS

- Increase the dataset size by incorporating more real-world accident data to improve analysis accuracy
- Apply advanced machine learning algorithms such as Random Forest, Gradient Boosting, or XGBoost for better prediction performance
- Perform hyperparameter tuning to optimize model accuracy and reduce errors
- Include additional features such as traffic density, road type, and driver behavior for deeper analysis
- Implement real-time data integration to monitor accident trends continuously
- Develop a user-friendly dashboard or web application for interactive visualization
- Enhance data visualization using advanced techniques like interactive charts and geospatial mapping
- Integrate a prediction system to forecast accident-prone areas and time periods

Git Hub Link of the project along with the report	https://github.com/24saiaditi-ux/DA-MINI-PROJECT/upload/main
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